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Capstone Client Handoff: GSTD Modeling and 7-Day GSTD 5 Alert

Purpose of This Handoff

This folder contains the final client handoff materials for the capstone project. The package is organized so the client can review the final dataset, final modeling notebooks, demo materials, and supporting research trail without searching through exploratory files.

The project focused on two modeling tasks:

- Task A: classify the current crop growth stage, referred to as GSTD.
- Task B: create a 7-day alert model for whether an observation will enter GSTD 5 within the next 7 days.

Recommended Starting Point

- 01_Final_Report_and_Executive_Summary
- 03_Final_Notebooks/TaskA_Current_GSTD_Classification_FINAL.ipynb
- 03_Final_Notebooks/TaskB_7Day_GSTD5_Alert_FINAL.ipynb
- 05_Client_Demo_Materials/Client_Demo_TaskA_Current_GSTD_Classification_FINAL.html
- 05_Client_Demo_Materials/Client_Demo_TaskB_7Day_GSTD5_Alert_FINAL.html

Folder Guide

Folder	Purpose
00_README_START_HERE	Orientation files, run order, and client handoff guide.
01_Final_Report_and_Executive_Summary	Final written summary, report, and presentation materials.
02_Data	Final validated dataset, raw DSSAT outputs, and short documentation explaining dataset origin and derived modeling columns.
03_Final_Notebooks	Locked final technical notebooks for Task A and Task B.
04_Model_Results_and_Figures	Exported result tables and model output artifacts for review.
05_Client_Demo_Materials	Client-facing demo notebooks and exported HTML versions.
06_Appendix_Research	Parser notebook, dataset creation workflow, early-stage analysis, and supporting research trail.

Final Dataset

The final modeling dataset is named final_validated_dataset.csv and is included in 02_Data. The final notebooks look for this file either in the same folder as the notebook or in ../02_Data/.

The dataset was created by parsing DSSAT output files, cleaning the parsed tables, and merging them into one validated modeling file. The step-by-step parsing, merging, and validation workflow is documented in 06_Appendix_Research/Dataset_Creation_and_Validation_Workflow.ipynb. More detail is also provided in 02_Data/dataset_notes.pdf and 02_Data/derived_columns_notes.pdf.

Final Model Summary

Task A: Current GSTD Classification

The final Task A notebook shows that current GSTD classification is feasible using practical client-feasible variables. The strongest feasible version uses Client-6 variables with LAID and past-looking time-series features.

Task B: 7-Day GSTD 5 Alert

The final Task B notebook frames the alert as a Stage 1 early-warning problem. This avoids inflated performance from trivial negative cases and focuses the model on the part of the crop development process where prediction is operationally useful.

The core recommended Task B model is the Client-Feasible TS LightGBM model. The Client-6 + LAID + TS LightGBM model is an optional extension if LAID is reliably available. In the finalized run, the recommended Client-Feasible TS LightGBM model had a Best F1 operating threshold of 0.9984 on the forward-time holdout split.

Important Interpretation Notes

- The dataset is DSSAT/simulation-derived, so model results should be validated against real operational or field data before production deployment.
- Task B should not be evaluated over all rows without caution because many growth stages create trivial negatives or deterministic positives.
- RUNNO-based grouped validation helps avoid leakage across repeated observations from the same simulation run.
- Forward-time holdout results are especially important because they test whether the model generalizes to future years.
- The operating threshold is a business decision: lower thresholds catch more true transitions but create more false alerts; higher thresholds reduce false alerts but may miss more transitions.

Recommended Use of the Materials

- Use the PDF and HTML files for quick client review without running code.
- Use the final notebooks if the client wants to inspect or rerun the analysis.
- Use Dataset_Creation_and_Validation_Workflow.ipynb if the client wants to verify how the final validated CSV was produced from raw DSSAT outputs.
- Use appendix notebooks only for background context or to understand earlier modeling decisions.
- Do not treat appendix notebooks as final reported results unless explicitly noted.